Appendices

Early Diabetes Risk Screening: Technical Report · Team 3: Sumit Singh Jadaun (lead), Priyamvadha, Amiya
Code, notebooks and results: https://github.com/jadonsinghsumit/early-diabetes-risk-screening

These appendices give the full tables behind the main text, the team's division of work, how to reproduce every result, and a glossary. All figures are aggregate outputs from the repository; no survey rows are shown.

Appendix A · Team contributions

Ownership follows the roadmap stages and the repository's commit history. Every member reviewed the final report.

Member	Role	Main contributions
Sumit Singh Jadaun	Project lead; stages 1–4	Roadmap and research question; endpoint decision (prediabetes counted as positive); data ingestion and quality checks; cleaning policy, review flags and data dictionary; exploratory analysis; the 25 planned hypothesis tests; package, command-line tool, tests and CI set-up; stages 5–8 plan.
Priyamvadha	Stages 5–6	Stratified 80/20 split and cross-validation folds; leakage-safe preprocessing; class-weighting and repeated-profile checks; tuning and comparison of logistic regression, random forest and XGBoost; paper-checklist baseline; adjusted odds ratios and permutation importance.
Amiya	Stages 7–8	Locked screening threshold; single test-set evaluation, subgroup and missed-case analysis; Windows reproduction and portability fixes; stage notebooks; README; technical report and presentation.

Appendix B · Data dictionary

The 22 columns of the analysis table. `Diabetes_binary` is the main endpoint derived from the three-class `Diabetes_012` column (0 → 0; 1 and 2 → 1). Source: reports/uploaded-combined/data_dictionary.csv.

Column	Type	Range	Meaning and codes
`Diabetes_binary`	binary	0–1	0: no diabetes; 1: self-reported prediabetes or diabetes
`HighBP`	binary	0–1	Ever told blood pressure was high; 0 no, 1 yes
`HighChol`	binary	0–1	Ever told cholesterol was high; 0 no, 1 yes
`CholCheck`	binary	0–1	Cholesterol checked in the past five years; 0 no, 1 yes
`BMI`	numeric	12–98	Body mass index in kg/m²; source-derived observed bounds
`Smoker`	binary	0–1	Smoked at least 100 cigarettes in lifetime; 0 no, 1 yes
`Stroke`	binary	0–1	Ever told of a stroke; 0 no, 1 yes
`HeartDiseaseorAttack`	binary	0–1	Coronary heart disease or myocardial infarction; 0 no, 1 yes
`PhysActivity`	binary	0–1	Physical activity in past 30 days, excluding job; 0 no, 1 yes
`Fruits`	binary	0–1	Consumes fruit at least once daily; 0 no, 1 yes
`Veggies`	binary	0–1	Consumes vegetables at least once daily; 0 no, 1 yes
`HvyAlcoholConsump`	binary	0–1	Heavy drinking under source sex-specific thresholds; 0 no, 1 yes
`AnyHealthcare`	binary	0–1	Has healthcare coverage; 0 no, 1 yes
`NoDocbcCost`	binary	0–1	Could not see a doctor because of cost in past year; 0 no, 1 yes
`GenHlth`	ordinal	1–5	General health: 1 excellent, 2 very good, 3 good, 4 fair, 5 poor
`MentHlth`	numeric	0–30	Days of poor mental health in past 30 days
`PhysHlth`	numeric	0–30	Days of poor physical health in past 30 days

Column	Type	Range	Meaning and codes
`DiffWalk`	binary	0–1	Serious difficulty walking or climbing stairs; 0 no, 1 yes
`Sex`	binary	0–1	Source coding: 0 female; 1 male
`Age`	ordinal	1–13	Age bands: 1=18–24, 2=25–29, 3=30–34, 4=35–39, 5=40–44, 6=45–49, 7=50–54, 8=55–59, 9=60–64, 10=65–69, 11=70–74, 12=75–79, 13=80+
`Education`	ordinal	1–6	Education: 1 never attended/kindergarten, 2 grades 1–8, 3 grades 9–11, 4 high school/GED, 5 college 1–3 years, 6 college 4+ years
`Income`	ordinal	1–8	Household income: 1 <$10k, 2 $10k–<15k, 3 $15k–<20k, 4 $20k–<25k, 5 $25k–<35k, 6 $35k–<50k, 7 $50k–<75k, 8 ≥$75k

Appendix C · All 25 planned hypothesis tests (stage 4)

Main endpoint, all 253,680 respondents. For every test H_0 is no association between the answer and the outcome. OR = unadjusted odds ratio (code 1 vs 0) with 95% CI; r_{rb} = rank-biserial correlation (positive means higher values among people with the condition); V = Cramér's V. p-values are Benjamini–Hochberg adjusted; values printed as < 1e-300 underflow to zero in double precision. These are associations, not causes. Source: reports/uploaded-combined/hypotheses.csv.

Answer	Test	Statistic	Effect size (95% CI)	Adjusted p
HighBP	χ^2	1.85e+04	OR = 4.782 (4.668–4.898)	< 1e-300
HighChol	χ^2	1.12e+04	OR = 3.242 (3.169–3.315)	< 1e-300
CholCheck	χ^2	1.17e+03	OR = 5.868 (5.231–6.583)	5.3e-256
Smoker	χ^2	1e+03	OR = 1.411 (1.381–1.441)	2.8e-219
Stroke	χ^2	2.79e+03	OR = 2.968 (2.846–3.096)	< 1e-300
HeartDiseaseorAttack	χ^2	7.94e+03	OR = 3.511 (3.411–3.614)	< 1e-300
PhysActivity	χ^2	3.74e+03	OR = 0.496 (0.484–0.507)	< 1e-300
Fruits	χ^2	449	OR = 0.790 (0.773–0.808)	1.1e-99
Veggies	χ^2	890	OR = 0.680 (0.663–0.698)	2.3e-195
HvyAlcoholConsump	χ^2	815	OR = 0.405 (0.380–0.432)	3.6e-179
AnyHealthcare	χ^2	50.3	OR = 1.209 (1.147–1.275)	1.3e-12
NoDocbcCost	χ^2	367	OR = 1.408 (1.360–1.459)	1.0e-81
DiffWalk	χ^2	1.25e+04	OR = 3.696 (3.608–3.785)	< 1e-300
Sex	χ^2	222	OR = 1.177 (1.152–1.202)	2.9e-50
BMI	Mann–Whitney U	5.86e+09	r_{rb} = 0.372 (0.366–0.377)	< 1e-300
GenHlth	χ^2	2.38e+04	V = 0.306	< 1e-300
GenHlth	Mann–Whitney U	6.19e+09	r_{rb} = 0.450 (0.445–0.455)	< 1e-300
MentHlth	Mann–Whitney U	4.52e+09	r_{rb} = 0.059 (0.053–0.064)	3.7e-115
PhysHlth	Mann–Whitney U	5.21e+09	r_{rb} = 0.221 (0.215–0.226)	< 1e-300
Age	χ^2	9.54e+03	V = 0.194	< 1e-300
Age	Mann–Whitney U	5.52e+09	r_{rb} = 0.293 (0.288–0.298)	< 1e-300
Education	χ^2	4.54e+03	V = 0.134	< 1e-300
Education	Mann–Whitney U	3.46e+09	r_{rb} = -0.189 (-0.195–-0.183)	< 1e-300

Answer	Test	Statistic	Effect size (95% CI)	Adjusted p
Income	χ^2	7.78e+03	V = 0.175	< 1e-300
Income	Mann–Whitney U	3.14e+09	r_{rb} = -0.266 (-0.271–-0.260)	< 1e-300

Appendix D · Model comparison on validation folds (stage 6)

Main endpoint, 202,944 training rows; mean ± SD over the five stratified validation folds of the tuned models. Source: reports/modeling/combined/training/comparison.csv.

Model	PR-AUC	ROC-AUC	Brier	Precision at 80% recall	Share flagged at 80% recall
No model	0.158 ± 0.000	0.500 ± 0.000	0.133	0.158	100.0%
Paper checklist (NIDDK/ADA)	0.325 ± 0.004	0.756 ± 0.004	0.228	0.256	51.4%
Logistic regression	0.439 ± 0.007	0.819 ± 0.004	0.108	0.324	38.9%
Random forest	0.456 ± 0.007	0.822 ± 0.004	0.106	0.326	38.7%
XGBoost (selected)	0.461 ± 0.006	0.826 ± 0.003	0.106	0.332	38.0%

Paired fold-by-fold comparisons

Nadeau–Bengio corrected resampled t-test on PR-AUC. Source: reports/modeling/combined/training/paired_differences.csv.

Model	Compared with	Mean PR-AUC gain	Folds won	Corrected t	p
Logistic regression	Paper checklist (NIDDK/ADA)	+0.114	5 of 5	38.0	2.9e-06
Random forest	Paper checklist (NIDDK/ADA)	+0.131	5 of 5	62.8	3.8e-07
XGBoost	Paper checklist (NIDDK/ADA)	+0.136	5 of 5	59.6	4.7e-07
Random forest	Logistic regression	+0.017	5 of 5	10.7	4.3e-04
XGBoost	Logistic regression	+0.022	5 of 5	15.3	1.1e-04
XGBoost	Random forest	+0.005	5 of 5	5.6	4.9e-03

Without the five healthcare-contact answers

Dropping `CholCheck`, `HighBP`, `HighChol`, `AnyHealthcare` and `NoDocbcCost`. Source: reports/modeling/combined/training/feature_sets.csv.

Model	PR-AUC, all 21 answers	PR-AUC, without contact answers	Change
Logistic regression	0.439	0.409	−0.030
Random forest	0.456	0.427	−0.029
XGBoost	0.461	0.431	−0.030

Appendix E · Test-set results by subgroup (stage 7)

XGBoost at the locked cutoff (0.153), 50,736 held-out respondents. Sensitivity intervals are 95% Wilson intervals. Source: reports/modeling/combined/evaluation/subgroups.csv.

Group	Level	People	With condition	Flagged	Sensitivity (95% CI)	False-alarm rate	Precision
Age	18-44	10,914	4.7%	7.8%	43.7% (39.5%–48.0%)	6.1%	26.4%
Age	45-64	22,230	15.5%	35.1%	76.3% (74.8%–77.6%)	27.5%	33.8%
Age	65+	17,592	22.9%	59.2%	86.6% (85.5%–87.6%)	51.0%	33.5%
Sex	Female	28,192	14.8%	34.6%	79.4% (78.1%–80.6%)	26.9%	33.9%
Sex	Male	22,544	17.0%	41.3%	79.4% (78.0%–80.6%)	33.5%	32.6%
Income	Under $25k	11,552	25.9%	61.1%	89.4% (88.2%–90.4%)	51.2%	37.8%
Income	$25k-$50k	12,458	17.3%	44.9%	81.1% (79.4%–82.7%)	37.3%	31.3%
Income	$50k+	26,726	10.7%	24.0%	67.6% (65.9%–69.3%)	18.8%	30.0%

Appendix F · Reproducing the results

Python 3.10 or later; exact library versions are pinned in requirements.lock. Download `archive.zip` from the Kaggle Diabetes Health Indicators dataset into `data/raw/`, then run:

```
python -m venv .venv && source .venv/bin/activate      # Windows: .venv\Scripts\activate
python -m pip install -r requirements.lock
python -m pip install --no-deps -e .

# Stages 1-4: prepare the endpoint, then ingestion, cleaning, exploration and tests
diabetes-screening prepare --input data/raw/archive.zip \
  --output data/derived/combined_BRFSS2015.csv --endpoint combined
diabetes-screening run --input data/derived/combined_BRFSS2015.csv \
  --provenance data/derived/combined_BRFSS2015.provenance.json --output outputs/stage1-4/combined

IN="--input data/derived/combined_BRFSS2015.csv --provenance data/derived/combined_BRFSS2015.provenance.json"
# Stage 5: split and imbalance checks
diabetes-screening-model split $IN --output outputs/combined/split
diabetes-screening-model imbalance $IN --split reports/modeling/combined/split --output
outputs/combined/imbalance
# Stage 6: tuning and comparison (20-25 minutes on a two-core laptop)
diabetes-screening-model train $IN --split reports/modeling/combined/split --output outputs/combined/training
# Stage 7: lock the cutoff on training rows, then score the test set once
diabetes-screening-model threshold $IN --split reports/modeling/combined/split \
  --training reports/modeling/combined/training --output outputs/combined/threshold
diabetes-screening-model test $IN --split reports/modeling/combined/split \
  --training reports/modeling/combined/training --threshold outputs/combined/threshold \
  --source data/raw/archive.zip --output outputs/combined/evaluation

python -m pytest      # 63 automated tests, fails below 80% coverage
```

For the prediabetes-only sensitivity analysis, use `--endpoint prediabetes` and replace `combined` with `prediabetes` in every path.

Where each result lives in the repository

Report section	Repository location
Stage 1 · quality checks and provenance	reports/uploaded-combined/quality.json, reports/uploaded-combined/source_provenance.json
Stage 2 · cleaning decisions and review flags	reports/uploaded-combined/cleaning.json, docs/cleaning.md
Stage 3 · figures 1–3 and observations	reports/uploaded-combined/eda/, docs/exploration.md
Stage 4 · hypothesis table (Appendix C)	reports/uploaded-combined/hypotheses.csv, docs/statistics.md
Stage 5 · split and imbalance checks	reports/modeling/combined/split/split.json, reports/modeling/combined/imbalance/README.md, docs/splitting.md
Stage 6 · tuning, comparison, odds ratios	reports/modeling/combined/training/README.md, docs/training.md
Stage 7 · threshold and test results	reports/modeling/combined/threshold/threshold.json, reports/modeling/combined/evaluation/README.md, docs/evaluation.md
Prediabetes-only analysis	reports/modeling/prediabetes/, reports/prediabetes/README.md
Verification records	docs/validation.md, docs/modeling-validation.md
Stage notebooks (one per stage)	notebooks/

Appendix G · Glossary

Term	Plain-English meaning
Sensitivity (recall)	Of the people who have the condition, the share the model flags.
Precision	Of the people the model flags, the share who really have the condition.
Specificity	Of the people without the condition, the share the model correctly does not flag.
False-alarm rate	Of the people without the condition, the share the model flags (1 – specificity).
PR-AUC	How well a model ranks cases above non-cases, focused on the rare positive class. A random ranking scores the share of positives (0.158 here).
ROC-AUC	The chance that a randomly chosen case gets a higher risk than a randomly chosen non-case (0.5 = guessing, 1 = perfect).
F2 score	A single score combining recall and precision that counts recall twice as much, because missed cases matter more.
Brier score / calibration	How close predicted risks are to what actually happens. Well calibrated means a stated 30% risk is right for about 30 in 100 people.
Odds ratio (OR)	How much higher the odds of the condition are in one group than another; 1 means no difference. Unadjusted ORs compare one answer at a time; adjusted ORs hold the other answers fixed.
Benjamini–Hochberg correction	An adjustment of p-values that limits false discoveries when many tests are run together.
Stratified split / folds	Dividing the data so every part keeps the same share of positives (15.76%).
Data leakage	Information from the test data slipping into training, which makes results look better than they really are.
Class weighting	Telling a model to count each rare positive case more heavily during training.
Locked threshold	The risk cutoff chosen on training data and saved before the test set was opened, so the test result is an honest estimate.